Introduction (Type II - Big Data Analytics)

This project aims to build a cloud-based big data analytics platform to conduct analysis of customer behavior from an online retail store in the UK. As it is relevant to real-world business problems with significant commercial implications, big data analytics could help inspire businesses in ways that can contribute to their successes but also provide better personalized experiences for customers.

Customer behavior analysis helps to formulate data-driven decision-making of retail businesses. Insights and predictions gathered from analytical tasks including customer lifetime value prediction, customer churn prediction, customer segmentation, and product association analysis help businesses personalize marketing campaigns, optimize inventory management, improve customer loyalty, increase customer satisfaction, and build customer loyalty to increase revenue and build sustainable business models. Data-driven decision-making not only helps businesses grow but also can improve the quality of life of customers and minimize environmental impact. Since this project enables highly targeted marketing through customer segmentation and predictive analytics, it helps businesses move away from wasteful mass-marketing practices.

The primary objective is to develop a scalable data pipeline using cloud computing to conduct data ingestion, processing, and analysis with Online Retail II dataset. The application will feature four analytical tasks, which consists of:

1. Customer Segmentation based on purchasing behavior (Clustering)
2. Customer Lifetime Value Prediction (Regression)
3. Customer Churn Prediction (Classification)
4. Product Association Analysis (Association Rule Mining)

Transaction records generate large amounts of data that traditional computing are unable to process efficiently. In this project, the dataset contains 1,067,371 observations, therefore performing analysis with Python or Excel at this scale may be slow or vulnerable to crashes. Having to train multiple machine learning models with traditional computing may also lack processing power and memory resources for computationally intensive tasks. Traditional computing also does not have built-in fault tolerance and distributed processing capabilities to reliably perform big data analytics.

A suitable method to address these constraints is by using Spark in cloud computing, an analytics engine capable of processing large scale data by distributing the workload across multiple nodes on a distributed cluster. Its scalability also allows data processing across multiple nodes that can significantly reduce computation time. Using cloud resources such as Google Cloud Platform also may be cheaper in comparison to adding physical hardware as supplementary computational resources. Services such as Spark and Hadoop Distributed Files System (HDFS) on docker containers help to simplify infrastructure management and improve accessibility to big data analytics services.

Technical Solutions

For data storage, the dataset will be stored in HDFS cluster and is deployed using Docker containers. Data processing and analytical tasks will be implemented by writing Spark programming using PySpark API with Spark MLlib library for machine learning. For development environment, the entire application will be developed and run from a Jupyter Notebook container for writing Spark MLlib algorithms, visualizing results, and documenting the process. All components will be deployed and managed using Docker Compose for straightforward orchestration and reproducibility of applications.

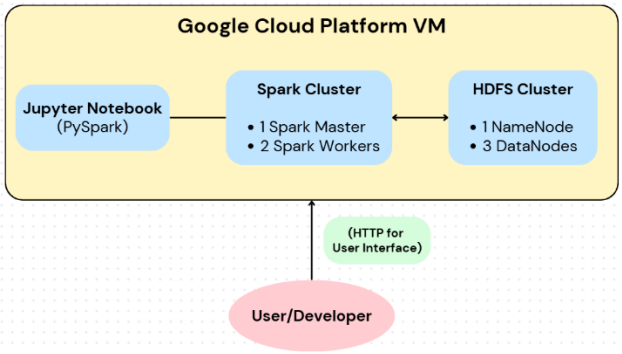
Monthly Cost Estimation (Budget Plan)

A machine configuration of n1-standard-2 costs $48.94 per month if powered on permanently. If the Virtual Machine (VM) is powered for a maximum of about 4 hours a day, and it is powered for this maximum time every day, then the VM will be powered for approximately 120 hours a month and the maximum cost breakdown will be:

1. Virtual Machine Instance Cost: ($48.94/ 720 hours) * 120 hours ≈ $8.16
2. Balanced persistent disk space of 100GB Cost: $4.00
3. Estimated Monthly Cost: US$12.16 ≈ AUD$18.24 (1.5 times conversion rate)

Since this cost is well within the $300 GCP credit, it is considered feasible and reasonable for the proposed cloud-based application.

Architecture Design



Workflow

1. Data stored in HDFS.
2. Jupyter Notebook connects to Spark Master.
3. Spark Workers read data from DataNodes.
4. Execute analytical tasks in Spark Cluster.
5. Show results in Jupyter Notebook.